

Nathan Zhao

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Education

Georgia Institute of Technology

Atlanta, Georgia

B.S. in Electrical Engineering (Circuit Technology, Robotics)

May 2029

Activities: Nanotechnology & Photonics Society · Propulsive Landers (Avionics)

The Pembroke Hill School

Kansas City, Missouri

High School Diploma, Cum Laude

ACT 36 · GPA 97.7 · Kevin R. Madden Science Award

Experience

Founding Engineer · Effluent

2026

- Designed and fabricated custom sensor-array carrier PCBs in KiCad, developed C++ microcontroller firmware interfacing with digital sensors over I2C and bridging telemetry to analytics systems with MQTT/open62541.
- Built simulation tools, Python data pipelines, and a cross-platform C#/.NET GUI/HMI delivering actionable ML scenario modeling insights directly to wastewater plant operators for enhanced process control.
- Engineered a machine learning model combining sparse attention and dilated inception GCNs for time-series forecasting of wastewater process data, focusing on transfer learning across facilities.

Research & Development Intern · MRIGlobal

2025 – 2026

- Led hardware development of portable, low-cost chemically-functionalized gas sensor arrays for real-time hazardous chemical classification, executing schematic capture and PCB layout in Altium Designer.
- Built sensor functionalization tools, a vacuum-oven vapor-deposition system, and sensor-characterization test rigs with closed-loop analyte mass-flow-control software via RS-232 communication and a custom Python HMI.
- Deployed optimized CNN-BiLSTM embedded models (TensorFlow Lite) on-device, achieving >96% preliminary classification accuracy on real-time edge inference for gas signatures.

Research Assistant · UMKC Multiplatform Interactive Robotics Lab (mirolab)

2025 – 2026

- Developed wearable hand-joint goniometry system using Kalman fusion of multi-view hand pose estimation with a custom FDM-printed ArUco glove, achieving sub-degree error in flexion-angle measurement validation.
- Designed the wearable's TPU kirigami/living-hinge flexures and a stepper-driven characterization rig for repeatable sensor validation; co-authoring manuscript intended for submission to *IEEE RA-L* (PI: Hee Sup Shin).

Research Intern · University of Pittsburgh, Accessible SmartTech Lab

2024

- Developed autonomous object-handoff behaviors on a Hello Robot Stretch RE1 (ROS Noetic) using ArUco markers, YOLOv5 object detection, and 2D LiDAR SLAM (PI: Dan Ding).

Selected Projects

Bimanual Mobile Manipulator for Assisted Living ↗

2024 – 2026

Regeneron ISEF 2025 4th Grand Award, GKCSEF 2025 Grand Winner

- Designed and built a dual-telescopic-arm mobile robot with custom 3D-printed wave-drive actuators, electromechanical architecture, power distribution, and embedded firmware.
- Implemented PIDF actuator control with quadrature encoder feedback for millimeter-scale arm positioning.

VIGIL: Assistive Eyewear & Autonomous Aerial Scout ↗

2026

HackGT 13 Best Overall: 3rd Place

- Co-designed and assembled assistive eyewear integrating an ESP32-CAM, Nano ESP32, dual VL53L0X ToF LiDAR sensors, servo-driven haptic pads, and USB-C power modules.
- Architected and developed dual-MCU firmware for video streaming, UDP telemetry, and haptic control; decoupled video/control paths and implemented nonblocking haptic patterns and command-loss failsafes.

Relevant Skills

Computer Aided Design: KiCad, Altium Designer, EasyEDA, Onshape, SolidWorks

Lab and Fabrication: Digital Multimeter, Oscilloscope, Logic Analyzer, Soldering (Through-hole and SMT), Probe Station, Physical Vapor Deposition, FDM/SLA 3D Printing

Embedded & Firmware: C/C++, PlatformIO, PicoSDK, Arduino IDE, I2C, SPI, UART, ROS/ROS2

Software and Machine Learning: Python, C#/.NET, Java, MATLAB, PyTorch, OpenCV, Linux, Docker, Git